Chapter 5. Ideals and Factor Rings

Ring homomorphisms, factor rings, prime and maximal ideals, Gröbner bases

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5.1 Homomorphisms and factor rings (§26)

Motivation. This section repeats the group-theoretic story of Chapter 3 for rings: kernels of ring homomorphisms are ideals, factor rings R/N exist exactly for ideals N, and every homomorphic image is a factor ring. $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ is the model example.

Theorem 5.1 (26.3)

Let $\phi : R \to R'$ be a ring homomorphism.
(1) $\phi(0) = 0'$ and $\phi(-a) = -\phi(a)$.
(2) If S is a subring of R, $\phi[S]$ is a subring of R'; if S' is a subring of R', $\phi^{-1}[S']$ is a subring of R.
(3) If R has unity 1, then $\phi(1)$ is unity for $\phi[R]$ (if $\phi[R] \neq \{0'\}$).

Proof. (1) and the additive parts of (2) are Theorem 3.1 for $\langle R, + \rangle$. Closure under multiplication: $\phi(a)\phi(b) = \phi(ab)$, and $\phi(a), \phi(b) \in S'$ gives $\phi(ab) \in S'$.

(3) $\phi(1)\phi(a) = \phi(a) = \phi(a)\phi(1)$. ■

$\ker \phi = \phi^{-1}[\{0'\}]$.

Theorem 5.2 (26.5)

For $H = \ker \phi$ and $a \in R$, $\phi^{-1}[\{\phi(a)\}] = a + H = H + a$.

Proof. Theorem 3.2 for the additive groups. ■

Corollary 5.3 (26.6)

A ring homomorphism is one-to-one iff its kernel is $\{0\}$.

Theorem 5.4 (26.7)

If $\phi : R \to R'$ is a ring homomorphism with kernel H, the additive cosets of H form a ring R/H with $(a+H)+(b+H) = (a+b)+H$ and $(a+H)(b+H) = (ab)+H$, and $\mu(a+H) = \phi(a)$ is an isomorphism $R/H \to \phi[R]$.

Proof. As in Theorem 3.5: μ is a bijection from cosets onto $\phi[R]$, and the operations are transported from $\phi[R]$ ($\phi(a)\phi(b) = \phi(ab)$), so they are well defined and make R/H a ring isomorphic to $\phi[R]$. ■

Theorem 5.5 (26.9)

Let H be a subring of R. Multiplication of additive cosets $(a + H)(b + H) = ab + H$ is well defined iff $ah \in H$ and $hb \in H$ for all $a, b \in R$, $h \in H$.

Proof. If it is well defined, compare the representatives a, 0 and a, h of $a+H$, $0+H$: $a0+H = ah+H$, so $ah \in H$; similarly $hb \in H$. Conversely, if $a' = a+h_1$, $b' = b+h_2$, then $a'b' = ab+ah_2+h_1b+h_1h_2 \in ab + H$. ■

An additive subgroup N with $aN \subset N$ and $Nb \subset N$ for all $a, b \in R$ is an **ideal**.

Corollary 5.6 (26.14)

If N is an ideal of R, the additive cosets of N form a ring R/N, the **factor ring**, under $(a+N)+(b+N)=(a+b)+N$ and $(a+N)(b+N)=ab+N$.

Proof. The operations are well defined (Corollary 3.7 and Theorem 5.5); the ring axioms follow from those of R. ■

Theorem 5.7 (26.16)

If N is an ideal, $\gamma : R \to R/N$, $\gamma(x) = x+N$, is a ring homomorphism with kernel N.

Theorem 5.8 (26.17, fundamental homomorphism theorem)

If $\phi : R \to R'$ is a ring homomorphism with kernel N, then $\phi[R]$ is a ring and $\mu(x+N)=\phi(x)$ is an isomorphism $R/N \to \phi[R]$ with $\phi = \mu \circ \gamma$.

Proof. Theorem 5.4; the kernel is an ideal, since $\phi(ah)=\phi(a)0'=0'$ and $\phi(hb)=0'$. ■

Example 5.1

$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$. The evaluation homomorphism $\phi_i : \mathbb{R}[x] \to \mathbb{C}$ has kernel $\langle x^2+1\rangle$ and image $\mathbb{C}$, so $\mathbb{R}[x]/\langle x^2+1\rangle \cong \mathbb{C}$.

5.2 Prime and maximal ideals (§27)

Motivation. The factor ring R/N inherits properties from N: it is a field exactly when N is maximal and an integral domain exactly when N is prime. This is the key to constructing fields: $F[x]/\langle p(x)\rangle$ is a field containing a root of p whenever p is irreducible, the starting point of Chapter 6.

Theorem 5.9 (27.5)

If R has unity and an ideal N contains a unit, then $N = R$.

Proof. If $u \in N$ is a unit, $1 = u^{-1}u \in N$, so $r = r1 \in N$ for all r. ■

Corollary 5.10 (27.6)

A field has no proper nontrivial ideals.

A **maximal ideal** is a proper ideal M not contained in any strictly larger proper ideal.

Theorem 5.11 (27.9)

Let R be commutative with unity. M is a maximal ideal iff R/M is a field.

Proof. ($\Rightarrow$) R/M is commutative with unity $1+M \neq 0+M$. For $a \notin M$, $N = \{ra+m : r \in R, m \in M\}$ is an ideal strictly containing M, so $N = R$ and $1 = ra+m$; then $(r+M)(a+M) = 1+M$.

($\Leftarrow$) If $M \subsetneq N$ with N an ideal, pick $a \in N - M$; $a+M$ has an inverse $b+M$, so $1-ab \in M \subset N$ and $1 = (1-ab)+ab \in N$, hence $N = R$. And $M \neq R$ since R/M is a field. ■

Corollary 5.12 (27.11)

A commutative ring with unity is a field iff it has no proper nontrivial ideals.

Proof. Corollary 5.10; conversely $\{0\}$ is maximal and $R \cong R/\{0\}$. ■

An ideal $N \neq R$ of a commutative ring is **prime** if $ab \in N$ implies $a \in N$ or $b \in N$.

Theorem 5.13 (27.15)

Let R be commutative with unity and $N \neq R$ an ideal. R/N is an integral domain iff N is prime.

Proof. $(a+N)(b+N) = N$ iff $ab \in N$, and $a+N = N$ iff $a \in N$; so "no divisors of 0 in R/N" is exactly primality. ■

Corollary 5.14 (27.16)

Every maximal ideal of a commutative ring with unity is prime.

Proof. A field is an integral domain. ■

Theorem 5.15 (27.17)

If R has unity, $\phi : \mathbb{Z} \to R$, $\phi(n) = n \cdot 1$, is a ring homomorphism.

Proof. $(n+m)\cdot 1 = n\cdot 1 + m\cdot 1$ and $(nm)\cdot 1 = (n\cdot 1)(m\cdot 1)$ by the distributive laws. ■

Corollary 5.16 (27.18)

If R has unity and characteristic $n > 1$, R contains a subring isomorphic to $\mathbb{Z}_n$; if characteristic 0, a subring isomorphic to $\mathbb{Z}$.

Proof. $\ker \phi = n\mathbb{Z}$; apply Theorem 5.8. ■

Theorem 5.17 (27.19)

A field F either has prime characteristic p and contains a subfield isomorphic to $\mathbb{Z}_p$, or has characteristic 0 and contains a subfield isomorphic to $\mathbb{Q}$.

Proof. The characteristic of an integral domain is 0 or prime. In the first case use Corollary 5.16. In the second, F contains a copy of $\mathbb{Z}$ and hence of its field of quotients $\mathbb{Q}$ (Corollary 4.22). ■

These subfields are the **prime fields**. An ideal $\langle a \rangle = \{ra : r \in R\}$ of a commutative ring with unity is **principal**.

Theorem 5.18 (27.24)

If F is a field, every ideal of $F[x]$ is principal.

Proof. Let $N \neq \{0\}$ and $g \in N$ nonzero of minimal degree. For $f \in N$, $f = gq + r$ with $r = 0$ or $\deg r < \deg g$; $r = f - gq \in N$, so $r = 0$. Hence $N = \langle g \rangle$. ■

Theorem 5.19 (27.25)

An ideal $\langle p(x)\rangle \neq \{0\}$ of $F[x]$ is maximal iff $p(x)$ is irreducible over F.

Proof. If $p = fg$ with lower degrees, then $\langle p\rangle \subsetneq \langle f\rangle \subsetneq F[x]$. Conversely, if p is irreducible and $\langle p\rangle \subset N = \langle g\rangle$, then $p = gq$, so g is a unit (then $N = F[x]$) or q is a unit (then $N = \langle p\rangle$). $\langle p\rangle \neq F[x]$ since p is not a unit. ■

So $F[x]/\langle p(x)\rangle$ is a field for irreducible p; for example $\mathbb{Z}_2[x]/\langle x^2 + x + 1\rangle$ is a field of 4 elements.

Theorem 5.20 (27.27)

If $p(x)$ is irreducible in $F[x]$ and divides $r(x)s(x)$, then it divides $r(x)$ or $s(x)$.

Proof. $\langle p\rangle$ is maximal, hence prime (Corollary 5.14), and $rs \in \langle p\rangle$. ■

5.3 Gröbner bases for ideals (§28)

Motivation. How do we decide whether a polynomial in several variables lies in a given ideal, or solve a system of polynomial equations? In one variable the division algorithm answers this; Gröbner bases are generating sets for which multivariable division behaves just as well. The Hilbert basis theorem guarantees that finite generating sets always exist.

For $S \subset F[x_1, \ldots, x_n]$ the **algebraic variety** $V(S) \subset F^n$ is the set of common zeros of the polynomials in S. A **basis** of an ideal I is a finite set $\{b_1, \ldots, b_r\} \subset I$ with $I = \langle b_1, \ldots, b_r\rangle = \{\sum f_i b_i\}$.

Theorem 5.21 (28.4)

The common zeros in F^n of $f_1, \ldots, f_r$ are the common zeros of all polynomials in $I = \langle f_1, \ldots, f_r\rangle$.

Proof. A common zero a of the f_i is a zero of every $\sum g_i f_i$, since evaluation at a is a ring homomorphism; the converse is clear as $f_i \in I$. ■

Theorem 5.22 (28.5, Hilbert basis theorem)

Every ideal of $F[x_1, \ldots, x_n]$ has a finite basis.

Proof [Supplement]. Call a commutative ring with unity *Noetherian* if all its ideals are finitely generated; fields are (Corollary 5.10). We show that R Noetherian implies $R[x]$ Noetherian; induction on n then gives the theorem. Suppose $I \subset R[x]$ is not finitely generated. Choose $f_1 \in I$ of least degree, and inductively $f_{k+1} \in I - \langle f_1, \ldots, f_k\rangle$ of least degree; then $d_k = \deg f_k$ is nondecreasing. Let a_k be the leading coefficient of f_k. The ideal $\langle a_1, a_2, \ldots\rangle$ of R is finitely generated, hence equals $\langle a_1, \ldots, a_N\rangle$ for some N; write $a_{N+1} = \sum_{i\le N} r_i a_i$. Then $g = f_{N+1} - \sum_{i\le N} r_i x^{d_{N+1}-d_i} f_i$ lies in $I - \langle f_1, \ldots, f_N\rangle$ and has degree $< d_{N+1}$, contradicting the choice of f_{N+1}. ■

A **power product** is $x_1^{m_1} \cdots x_n^{m_n}$. A **term order** $<$ is a total order of power products with $1 < P$ for $P \neq 1$ and $P < Q \Rightarrow PR < QR$; e.g. the lexicographic order with $x_1 > x_2 > \cdots > x_n$. It is a well-ordering (no infinite strictly decreasing sequence), by Dickson's lemma: every set of power products has finitely many minimal elements under divisibility (a consequence of Theorem 5.22 applied to the monomial ideal they generate). For $f \neq 0$ write $\mathrm{lp}(f)$ for its largest power product and $\mathrm{lt}(f)$ for its leading term. Division by $g_1, \ldots, g_r$ (repeatedly cancel the largest term divisible

by some $\operatorname{lp}(g_i)$) writes $f = \sum q_i g_i + r$ with $\operatorname{lp}(q_i g_i) \le \operatorname{lp}(f)$ and no term of r divisible by any $\operatorname{lp}(g_i)$; it terminates by the well-ordering.

Theorem 5.23 (28.6, property of the division algorithm)

If $f = gq + r$ in $F[\mathbf{x}]$, the common zeros of f, g are those of g, r, and the common divisors of f, g are those of g, r. Hence replacing f by r in a basis does not change the ideal.

Proof. $f \in \langle g, r\rangle$ and $r = f - gq \in \langle f, g\rangle$, so $\langle f, g\rangle = \langle g, r\rangle$; zeros and divisors follow. ■

$\{g_1, \dots, g_r\}$ is a **Gröbner basis** of $I = \langle g_1, \dots, g_r\rangle$ if for every nonzero $f \in I$ some $\operatorname{lp}(g_i)$ divides $\operatorname{lp}(f)$. For monic g_i, g_j with $L = \operatorname{lcm}(\operatorname{lp} g_i, \operatorname{lp} g_j)$, the **S-polynomial** is $S(g_i, g_j) = \frac{L}{\operatorname{lt} g_i} g_i - \frac{L}{\operatorname{lt} g_j} g_j$, in which the leading terms cancel.

Theorem 5.24 (28.12, Buchberger's criterion)

$G = \{g_1, \dots, g_r\}$ is a Gröbner basis of $\langle G\rangle$ iff every $S(g_i, g_j)$ reduces to 0 by repeated division by elements of G.

Proof [Supplement]. We may assume the g_i monic.

($\Rightarrow$) $S(g_i, g_j) \in I$; its remainder r on division lies in I and has no term divisible by any $\operatorname{lp}(g_k)$; if $r \neq 0$ this contradicts the Gröbner property for $\operatorname{lp}(r)$.

($\Leftarrow$) Reduction to 0 gives a representation $S(g_i, g_j) = \sum_k q_k g_k$ with $\operatorname{lp}(q_k g_k) \le \operatorname{lp} S(g_i, g_j) < L_{ij}$. Let $f \in I$ be nonzero. Among all representations $f = \sum h_i g_i$ choose one with $\delta = \max_i \operatorname{lp}(h_i g_i)$ minimal (well-ordering). Then $\operatorname{lp}(f) \le \delta$; if equality holds, $\operatorname{lp}(f) = \operatorname{lp}(h_i)\operatorname{lp}(g_i)$ for some i, and we are done. Suppose $\operatorname{lp}(f) < \delta$, and let T be the set of i with $\operatorname{lp}(h_i g_i) = \delta$, $\operatorname{lt}(h_i) = c_i P_i$. The terms in δ cancel, so $\sum_{i\in T} c_i = 0$. Put $p_i = P_i g_i$ (monic, $\operatorname{lp} = \delta$) and order $T = \{i_1, \dots, i_s\}$. Then

$$\sum_{i\in T} c_i p_i = \sum_{k<s} (c_{i_1} + \cdots + c_{i_k})(p_{i_k} - p_{i_{k+1}}),$$

and $p_{i_k} - p_{i_{k+1}} = \frac{\delta}{L} S(g_{i_k}, g_{i_{k+1}})$ with $L = L_{i_k i_{k+1}}$, which by hypothesis is $\sum_m q'_m g_m$ with all $\operatorname{lp}(q'_m g_m) < \delta$. Substituting into $f = \sum_{i\in T} c_i p_i + \sum_{i\in T}(h_i - c_i P_i) g_i + \sum_{i\notin T} h_i g_i$ gives a representation with maximum $< \delta$, a contradiction. ■

Buchberger's algorithm: add to G the nonzero remainders of the S-polynomials until all reduce to 0; it terminates because the ideal generated by the leading power products strictly grows and ascending chains of ideals stabilize (Theorem 5.22). Gaussian elimination and the Euclidean algorithm are the linear and one-variable cases.

Example 5.2

For $I = \langle xy, y^2 - y\rangle$ in $\mathbb{R}[x, y]$ (lex, $x > y$): $S(xy, y^2 - y) = y \cdot xy - x(y^2 - y) = xy$, which reduces to 0. So $\{xy, y^2 - y\}$ is a Gröbner basis, and $V(I) = \{(x, 0)\} \cup \{(0, 1)\}$.